Summary: From DFT to FFT and Optical FFT Hardware

Part I. DFT $\to$ FFT (Core Derivation, Before 9.3.1)

Step 1: Start from the N-point DFT

According to the definition of the discrete Fourier transform (DFT),

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}, \qquad W_N = e^{-j2\pi/N}, \qquad k = 0, 1, \ldots, N-1.$$

For direct computation, each output needs about N multiplications and $N-1$ additions, so the total cost is $\mathcal{O}(N^2)$.

Step 2: Split into even and odd indices

Before rewriting the summation, define the even and odd subsequences:

$$x_{\text{even}}[r] = x[2r], \qquad x_{\text{odd}}[r] = x[2r+1], \qquad r = 0, 1, \ldots, \frac{N}{2} - 1.$$

Then write $n = 2r$ and $n = 2r+1$:

$$\begin{aligned} X[k] &= \sum_{r=0}^{N/2-1} x_{\text{even}}[r] W_N^{2rk} + \sum_{r=0}^{N/2-1} x_{\text{odd}}[r] W_N^{(2r+1)k} \\ &= \sum_{r=0}^{N/2-1} x[2r] W_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1] W_N^{(2r+1)k} \\ &= \sum_{r=0}^{N/2-1} x[2r] (W_{N/2})^{rk} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1] (W_{N/2})^{rk}. \end{aligned}$$

Define two $(N/2)$-point DFTs:

$$G[k] = \sum_{r=0}^{N/2-1} x[2r] (W_{N/2})^{rk}, \qquad H[k] = \sum_{r=0}^{N/2-1} x[2r+1] (W_{N/2})^{rk}.$$

Then

$$X[k] = G[k] + W_N^k H[k].$$

Step 3: Use periodicity to get output pairs

Because $G[k]$ and $H[k]$ are length-$(N/2)$ DFT outputs, they are periodic with period $N/2$:

$$G[k + N/2] = G[k], \qquad H[k + N/2] = H[k].$$

Also,

$$W_N^{k+N/2} = W_N^k W_N^{N/2} = -W_N^k.$$

Hence, for $k = 0, 1, \ldots, N/2 - 1$:

$$X[k] = G[k] + W_N^k H[k],$$
$$X[k + N/2] = G[k] - W_N^k H[k].$$

since $W_N^{r+N/2} = -W_N^r$, we only need to load one coefficient instead of two. The refined diagram is shown in Fig. 1, where the coefficient is multiplied by the lower-arm input first, and then added to and subtracted from the upper-arm input to produce two outputs. So each pair $\{X[k], X[k+N/2]\}$ is produced by one butterfly fed by the same $G[k], H[k]$. Then this $(N/2)$ DFT can be further decomposed as $(N/4)$ DFT, $(N/8)$ DFT and finally, 2 point DFT. The final diagram is shown in Fig. 1.

For a 2-point input $[a_0, a_1]^T$,

$$X_2[0] = a_0 + a_1, \qquad X_2[1] = a_0 - a_1,$$

or in matrix form,

$$\begin{bmatrix} X_2[0] \\ X_2[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}.$$

where a_0, a_1 are the input of FFT.

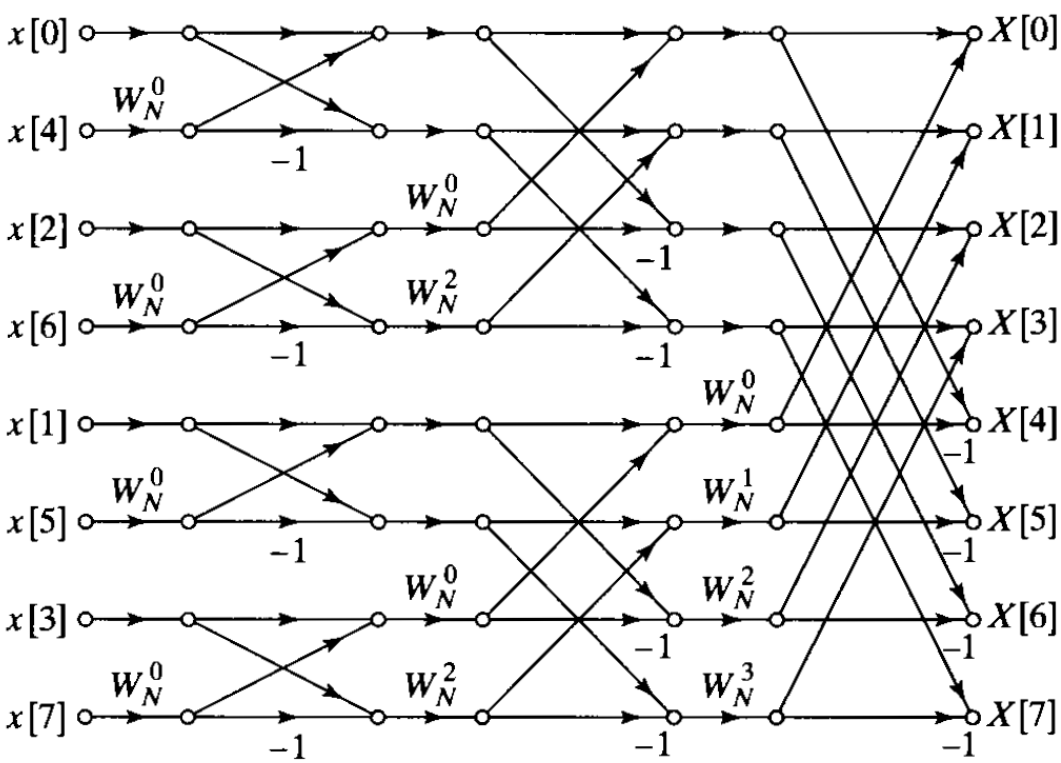

Figure 1: Overall flow of FFT based on butterfly structure

Complexity summary

From the simplified-butterfly derivation above, each stage has $N/2$ butterflies, and each butterfly uses one complex multiplication and two additions. With $\log_2 N$ stages:

$$\text{Multiplications} = \frac{N}{2}\log_2 N, \qquad \text{Additions} = N\log_2 N.$$

Compared with direct DFT (about N^2 multiplications and $N(N-1)$ additions), this is the FFT reduction to $\mathcal{O}(N\log_2 N)$.

Part II. Computing FFT using Photonic Integrated Circuit

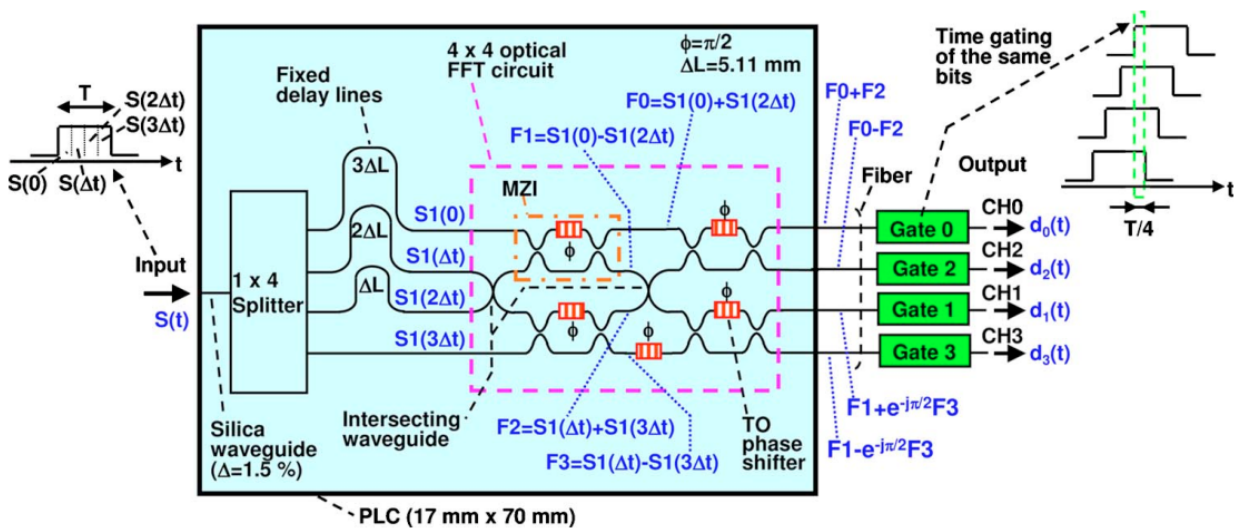

Figure 2: 4-point optical FFT block used in the OFDM demultiplexer (cropped from the paper figure).

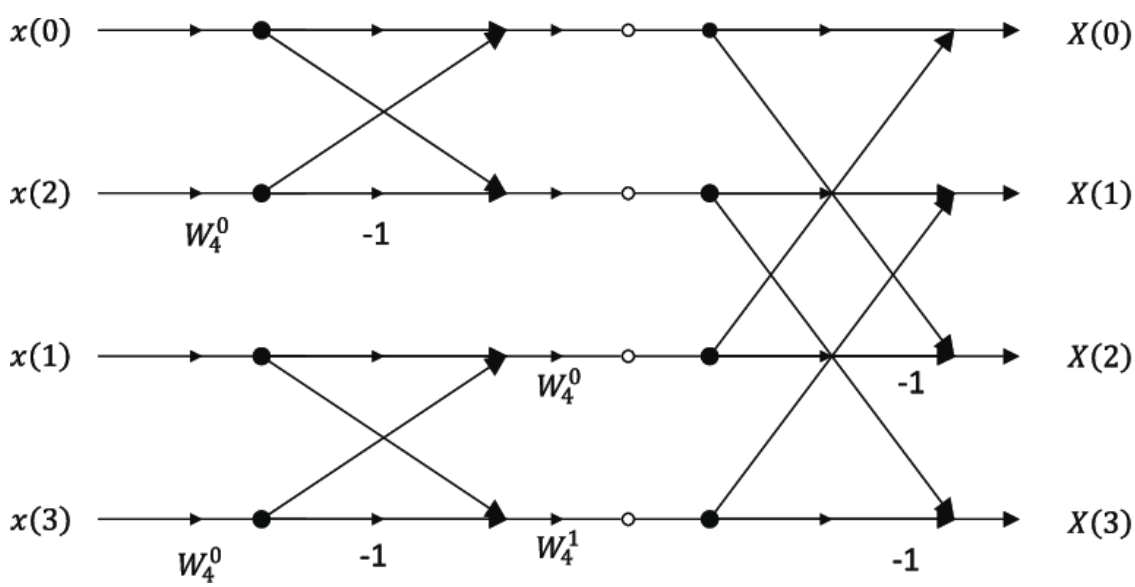

Figure 3: diagram of 4 point FFT.

2.1 4-point FFT equations to implement

Let

$$x_0 = S_1(0), \quad x_1 = S_1(\Delta t), \quad x_2 = S_1(2\Delta t), \quad x_3 = S_1(3\Delta t),$$

and $W_4 = e^{-j\pi/2} = -j$.

First stage (two butterflies):

$$F_0 = x_0 + x_2, \quad F_1 = x_0 - x_2, \qquad F_2 = x_1 + x_3, \quad F_3 = x_1 - x_3.$$

Second stage:

$$d_0 = F_0 + F_2, \quad d_2 = F_0 - F_2, \quad d_1 = F_1 + e^{-j\pi/2} F_3, \quad d_3 = F_1 - e^{-j\pi/2} F_3.$$

So only one non-trivial twiddle is needed: $e^{-j\pi/2}$ on the F_3 branch.

2.2 Why each MZI realizes a butterfly

For the actual device shown in the figure, each symmetric MZI appears to use a single thermo-optic (TO) phase shifter placed on one arm. If the heated arm is the upper arm and the TO phase shifter is modeled as a negative phase shift, then the physical arm propagation matrix is

$$P_{\text{phys}}(\phi) = \begin{bmatrix} e^{-j\phi} & 0 \\ 0 & 1 \end{bmatrix}.$$

Using the paper-consistent 3 dB coupler convention,

$$C = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix},$$

the physical MZI transfer matrix is

$$T_{\text{phys}}(\phi) = C\, P_{\text{phys}}(\phi)\, C.$$

Substituting the matrices gives

$$P_{\text{phys}}(\phi)\, C = \begin{bmatrix} e^{-j\phi} & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} e^{-j\phi} & -je^{-j\phi} \\ -j & 1 \end{bmatrix},$$

and therefore

$$T_{\text{phys}}(\phi) = \frac{1}{2} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} e^{-j\phi} & -je^{-j\phi} \\ -j & 1 \end{bmatrix}.$$

Carrying out the multiplication yields

$$T_{\text{phys}}(\phi) = \frac{1}{2} \begin{bmatrix} e^{-j\phi} - 1 & -j\left(e^{-j\phi} + 1\right) \\ -j\left(e^{-j\phi} + 1\right) & 1 - e^{-j\phi} \end{bmatrix}.$$

Now factor out the common phase term $e^{-j\phi/2}$. Using

$$e^{-j\phi} - 1 = -2j\, e^{-j\phi/2} \sin\left(\frac{\phi}{2}\right),$$

$$e^{-j\phi} + 1 = 2\, e^{-j\phi/2} \cos\left(\frac{\phi}{2}\right),$$

$$1 - e^{-j\phi} = 2j\, e^{-j\phi/2} \sin\left(\frac{\phi}{2}\right),$$

we obtain

$$T_{\text{phys}}(\phi) = -j\, e^{-j\phi/2} \begin{bmatrix} \sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) \\ \cos\left(\frac{\phi}{2}\right) & -\sin\left(\frac{\phi}{2}\right) \end{bmatrix}.$$

Thus, the physical single-heater model differs from the paper's printed MZI matrix only by a global phase factor:

$$T_{\text{phys}}(\phi) = e^{-j\phi/2}\, T_{\text{paper}}(\phi),$$

where

$$T_{\text{paper}}(\phi) = -j \begin{bmatrix} \sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) \\ \cos\left(\frac{\phi}{2}\right) & -\sin\left(\frac{\phi}{2}\right) \end{bmatrix}.$$

Therefore, the matrix printed in the paper should be understood as a phase-normalized form of the actual single-arm-heated MZI, rather than as implying that the two arms are physically driven with phase shifts $\pm\phi/2$.

At $\phi = \pi/2$,

$$T(\pi/2) = -\frac{j}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Ignoring the common factor $-j/\sqrt{2}$ (a global scaling), each MZI realizes an add/subtract butterfly.

2.3 One-to-one mapping: butterfly vs circuit blocks

Element	**FFT function in the 4-point flow**	**Phase setting**
MZI_1 (left-top)	Computes $F_0 = x_0 + x_2$ and $F_1 = x_0 - x_2$ (first-stage butterfly for even-index pair)	$\phi_1 = \pi/2$
MZI_2 (left-bottom)	Computes $F_2 = x_1 + x_3$ and $F_3 = x_1 - x_3$ (first-stage butterfly for odd-index pair)	$\phi_2 = \pi/2$
Crossing C_1 (input-side)	Permutes waveguides so the physical neighbors match butterfly pairs (x_0, x_2) and (x_1, x_3)	ideal 0 phase (routing only)
Crossing C_2 (inter-stage)	Permutes first-stage outputs so second-stage MZIs receive (F_0, F_2) and (F_1, F_3) pairings	ideal 0 phase (routing only)
TO twiddle shifter (on F_3 path)	Implements the required twiddle multiplication before the odd branch second-stage butterfly	$\theta_{\text{tw}} = -\pi/2$
MZI_3 (right-top)	Computes $d_0 = F_0 + F_2$ and $d_2 = F_0 - F_2$ (second-stage even branch)	$\phi_3 = \pi/2$
MZI_4 (right-bottom)	Computes $d_1 = F_1 + e^{-j\pi/2} F_3$ and $d_3 = F_1 - e^{-j\pi/2} F_3$ (second-stage odd branch)	$\phi_4 = \pi/2$